Towed Movable Antenna-Assisted Near-Field Integrated Sensing and Communication

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***Abstract*—Your abstract goes here.**

***Index Terms*—Towed movable antenna (ToMA), integrated sensing and communication (ISAC), near-field, full-duplex, antenna position optimization.**

I. Introduction

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II. System Model and Problem Formulation

Consider a ToMA-aided full-duplex near-field integrated sensing and communication (ISAC) system. The main UAV serves as the central platform for communication and sensing and is assumed to hover at a fixed location. It tows multiple UAVs through M fixed-length cables to form reconfigurable transmit and receive arrays. All cables are attached to a common anchor point on the main UAV, where half of the towing UAVs constitute the transmit ToMA array and the remaining half constitute the receive ToMA array. Over the same time-frequency resource, the transmit ToMA array simultaneously communicates with K downlink (DL) users and illuminates L sensing targets, while the receive ToMA array simultaneously receives communication signals from J uplink (UL) users and reflected echoes from multiple sensing targets. In this paper, the trajectory, global translation, and attitude of the main UAV are not optimized. Instead, the system performance is enhanced by jointly optimizing the ToMA geometry and the communication and sensing resources.

Unlike conventional fixed-position antenna (FPA) arrays and movable antenna (MA) architectures in which individual antenna elements can move independently, the ToMA array reconfigures its geometry by adjusting the positions of the towing UAVs, such that multiple antenna elements mounted on the same cable move in a coupled manner. Since the array aperture of the ToMA varies with the towing-UAV positions, the near-field operating region is determined according to the actual apertures of the transmit and receive ToMA arrays. Let D_{T} and D_{R} denote the effective apertures of the transmit and receive ToMA arrays, respectively. For $x \in \{\mathrm{T}, \mathrm{R}\}$, the corresponding radiative near-field region can be approximately characterized as $0.62\sqrt{D_x^3/\lambda} < r < 2D_x^2/\lambda$, where λ denotes the carrier wavelength. In this paper, the communication users and sensing targets are considered to lie within the radiative near-field region of the ToMA arrays. Accordingly, spherical-wave channel models are adopted for the communication and sensing links.

Moreover, to account for the self-interference (SI) caused by simultaneous transmission and reception in the full-duplex mode, we adopt a unified equivalent residual self-interference (RSI) model. Specifically, the aggregate residual effect after passive isolation, analog cancellation, and digital cancellation is represented by $\mathbf{H}_{\mathrm{RSI}}$. The RSI is not imposed as an additional feasibility constraint; instead, it affects the system weighted sum rate (WSR) through the SINRs of the UL communication and sensing links.

A. ToMA Geometry Model

To describe the three-dimensional (3D) geometry of the ToMA arrays, we establish a global 3D Cartesian coordinate system $O-xyz$ centered at the main UAV, with the common cable anchor point coinciding with the origin, i.e., $\mathbf{q}_0 = \mathbf{o} = \mathbf{0}$. Let the sets of DL users, UL users, and sensing targets be $\mathcal{K} = \{1, \ldots, K\}$, $\mathcal{J} = \{1, \ldots, J\}$, and $\mathcal{L} = \{1, \ldots, L\}$, respectively. Their 3D coordinates are denoted by $\mathbf{q}_k^{\mathrm{D}}$, $\mathbf{q}_j^{\mathrm{U}}$, and $\mathbf{q}_l^{\mathrm{S}} \in \mathbb{R}^3$, where $k \in \mathcal{K}$, $j \in \mathcal{J}$, and $l \in \mathcal{L}$.

The system contains M towing UAVs, where M is assumed to be even. Define the towing-UAV index set as $\mathcal{M} = \{1, \ldots, M\}$. The first $M/2$ UAVs form the transmit ToMA array, while the remaining $M/2$ UAVs form the receive ToMA array, i.e., $\mathcal{M}_{\mathrm{T}} = \{1, \ldots, M/2\}$ and $\mathcal{M}_{\mathrm{R}} = \{M/2 + 1, \ldots, M\}$. The transmit/receive role of each towing UAV remains fixed within one optimization interval. Each cable carries N_{c} uniformly distributed antenna elements. Hence, the transmit and receive arrays each contain $N_{\mathrm{T}} = N_{\mathrm{R}} = MN_{\mathrm{c}}/2$ antenna elements.

Let $\mathbf{c}_m \in \mathbb{R}^3$ denote the endpoint position of the m-th towing UAV relative to the main UAV. Since the antenna elements are uniformly distributed along the m-th cable, the position of its n-th antenna element is given by

$$\mathbf{p}_{m,n} = \frac{n}{N_{\mathrm{c}}}\mathbf{c}_m, \qquad n = 1, \ldots, N_{\mathrm{c}}. \tag{1}$$

Therefore, all antenna-element positions on the same cable are jointly determined by the corresponding endpoint $\mathbf{c}_m$.

For a transmit cable $m \in \mathcal{M}_{\mathrm{T}}$, define the global transmit-antenna index as $i = (m-1)N_{\mathrm{c}} + n$, and let $\mathbf{p}_i^{\mathrm{T}} := \mathbf{p}_{m,n}$, $i = 1, \ldots, N_{\mathrm{T}}$. Similarly, for a receive cable $m \in \mathcal{M}_{\mathrm{R}}$, define the global receive-antenna index as $s = \left(m - \frac{M}{2} - 1\right) N_{\mathrm{c}} + n$, and let $\mathbf{p}_s^{\mathrm{R}} := \mathbf{p}_{m,n}$, $s = 1, \ldots, N_{\mathrm{R}}$.

The endpoint positions associated with the transmit and receive ToMA arrays are stacked as $\widetilde{\mathbf{c}}_{\mathrm{T}} = \left[\mathbf{c}_m^T\right]_{m \in \mathcal{M}_{\mathrm{T}}}^T$ and

$\widetilde{\mathbf{c}}_{\mathrm{R}} = \left[\mathbf{c}_m^T\right]_{m\in\mathcal{M}_{\mathrm{R}}}^T$, respectively. The joint ToMA endpoint-position vector is further defined as $\widetilde{\mathbf{c}} = [\widetilde{\mathbf{c}}_{\mathrm{T}}^T, \widetilde{\mathbf{c}}_{\mathrm{R}}^T]^T$. Since the communication, sensing, and SI channels all vary with these endpoint positions, $\widetilde{\mathbf{c}}$ constitutes the key geometric optimization variable of the proposed ToMA system.

Since all cables have a fixed length L_{c}, the endpoint of each towing UAV must satisfy

$$\|\mathbf{c}_m\|_2 = L_{\mathrm{c}}, \qquad \forall m \in \mathcal{M}. \tag{2}$$

In addition, to avoid collisions among the towing UAVs, the endpoint positions are constrained by

$$\|\mathbf{c}_m - \mathbf{c}_{m'}\|_2 \geq D_{\mathrm{UAV}}, \qquad \forall m \neq m'. \tag{3}$$

Thus, each towing UAV is constrained to move on a sphere centered at the main UAV with radius L_{c}, while also satisfying the minimum inter-UAV safety-distance requirement.

B. Near-Field Channel Model

We consider a quasi-static narrowband line-of-sight (LoS) propagation environment and adopt the uniform spherical wave (USW) model for the communication, sensing, and SI channels. Unlike the conventional far-field plane-wave model, the near-field spherical-wave model preserves the actual propagation distance between each antenna element and each user or target, thereby capturing the phase variations induced by ToMA geometry reconfiguration. Let λ denote the carrier wavelength and define the wavenumber as $k_0 = 2\pi/\lambda$. For an arbitrary spatial position $\mathbf{q}$, the near-field array response vectors of the transmit and receive ToMA arrays are respectively defined as

$$\mathbf{a}_{\mathrm{T}}(\widetilde{\mathbf{c}}_{\mathrm{T}}, \mathbf{q}) = \left[e^{jk_0\|\mathbf{p}_i^{\mathrm{T}} - \mathbf{q}\|_2}\right]_{i=1}^{N_{\mathrm{T}}}, \tag{4}$$

and

$$\mathbf{a}_{\mathrm{R}}(\widetilde{\mathbf{c}}_{\mathrm{R}}, \mathbf{q}) = \left[e^{jk_0\|\mathbf{p}_s^{\mathrm{R}} - \mathbf{q}\|_2}\right]_{s=1}^{N_{\mathrm{R}}}. \tag{5}$$

Here, $\|\mathbf{p}_i^{\mathrm{T}} - \mathbf{q}\|_2$ and $\|\mathbf{p}_s^{\mathrm{R}} - \mathbf{q}\|_2$ represent the actual propagation distances from the i-th transmit antenna and the s-th receive antenna to position $\mathbf{q}$, respectively. Therefore, changing the towing-UAV endpoint positions alters the antenna coordinates and, consequently, the spherical-wave phase responses of the corresponding links.

Let $r_k^{\mathrm{D}} = \|\mathbf{q}_k^{\mathrm{D}}\|_2$ and $r_j^{\mathrm{U}} = \|\mathbf{q}_j^{\mathrm{U}}\|_2$ denote the reference propagation distances from the common anchor point at the main UAV to DL user k and UL user j, respectively. Under free-space LoS propagation, the corresponding common amplitude path gains are modeled as $\alpha_k^{\mathrm{D}} = \lambda/(4\pi r_k^{\mathrm{D}})$ and $\alpha_j^{\mathrm{U}} = \lambda/(4\pi r_j^{\mathrm{U}})$, respectively.

Accordingly, the DL channel of user k is expressed as

$$\mathbf{h}_k = \alpha_k^{\mathrm{D}} \mathbf{a}_{\mathrm{T}}(\widetilde{\mathbf{c}}_{\mathrm{T}}, \mathbf{q}_k^{\mathrm{D}}), \qquad k \in \mathcal{K}, \tag{6}$$

while the UL channel of user j is

$$\mathbf{f}_j = \alpha_j^{\mathrm{U}} \mathbf{a}_{\mathrm{R}}(\widetilde{\mathbf{c}}_{\mathrm{R}}, \mathbf{q}_j^{\mathrm{U}}), \qquad j \in \mathcal{J}. \tag{7}$$

Thus, the USW model absorbs the approximately common amplitude attenuation across the array into the link-specific path gain, while the element-wise exact propagation distances are retained in the array response vectors to characterize the near-field spherical-wave phases.

For sensing target l, since the transmit and receive ToMA arrays share the same anchor point at the origin, the reference propagation distances are given by $r_{\mathrm{T},l}^{\mathrm{S}} = \|\mathbf{q}_l^{\mathrm{S}}\|_2$ and $r_{\mathrm{R},l}^{\mathrm{S}} = \|\mathbf{q}_l^{\mathrm{S}}\|_2$. Accordingly, the one-way free-space power gains of the transmit-ToMA-to-target and target-to-receive-ToMA links are $\beta_{\mathrm{T},l}^{\mathrm{S}} = (\lambda/(4\pi r_{\mathrm{T},l}^{\mathrm{S}}))^2$ and $\beta_{\mathrm{R},l}^{\mathrm{S}} = (\lambda/(4\pi r_{\mathrm{R},l}^{\mathrm{S}}))^2$, respectively.

Let ξ_l^{S} denote the complex reflection coefficient of target l. The equivalent round-trip complex amplitude coefficient is then defined as

$$\alpha_l^{\mathrm{S}} = \xi_l^{\mathrm{S}} \sqrt{\beta_{\mathrm{T},l}^{\mathrm{S}} \beta_{\mathrm{R},l}^{\mathrm{S}}}. \tag{8}$$

Accordingly, the round-trip sensing channel matrix associated with target l is given by

$$\mathbf{G}_l = \alpha_l^{\mathrm{S}} \mathbf{a}_{\mathrm{R}}^*(\widetilde{\mathbf{c}}_{\mathrm{R}}, \mathbf{q}_l^{\mathrm{S}}) \mathbf{a}_{\mathrm{T}}^H(\widetilde{\mathbf{c}}_{\mathrm{T}}, \mathbf{q}_l^{\mathrm{S}}), \qquad l \in \mathcal{L}, \tag{9}$$

where $\mathbf{G}_l \in \mathbb{C}^{N_{\mathrm{R}} \times N_{\mathrm{T}}}$. The (s,i)-th entry contains the round-trip propagation phase from the i-th transmit antenna to the target and then from the target to the s-th receive antenna, i.e., $e^{-jk_0(\|\mathbf{p}_i^{\mathrm{T}} - \mathbf{q}_l^{\mathrm{S}}\|_2 + \|\mathbf{p}_s^{\mathrm{R}} - \mathbf{q}_l^{\mathrm{S}}\|_2)}$. Hence, the sensing channel depends jointly on the geometries of both the transmit and receive ToMA arrays.

We assume that the target location $\mathbf{q}_l^{\mathrm{S}}$ and the reflection coefficient ξ_l^{S} are available at the main UAV, either known a priori or estimated from previous sensing observations. Together with the distance-dependent propagation gains above, this enables the main UAV to construct $\mathbf{G}_l$ for the subsequent joint design of communication resources, sensing resources, and ToMA positions.

Since the main UAV operates in the full-duplex mode, the simultaneous operation of the transmit and receive ToMA arrays results in direct SI through the propagation paths between transmit and receive antenna elements. Define the distance between the i-th transmit antenna and the s-th receive antenna as $d_{s,i}^{\mathrm{SI}} = \|\mathbf{p}_s^{\mathrm{R}} - \mathbf{p}_i^{\mathrm{T}}\|_2$. Taking the actual Tx–Rx antenna distance into account, the (s,i)-th entry of the uncancelled direct SI channel $\mathbf{H}_{\mathrm{SI}}^0 \in \mathbb{C}^{N_{\mathrm{R}} \times N_{\mathrm{T}}}$ is modeled as

$$[\mathbf{H}_{\mathrm{SI}}^0]_{s,i} = \frac{\lambda}{4\pi d_{s,i}^{\mathrm{SI}}} e^{-jk_0 d_{s,i}^{\mathrm{SI}}}. \tag{10}$$

Here, the free-space amplitude attenuation is represented by $\lambda/(4\pi d_{s,i}^{\mathrm{SI}})$, while the propagation phase is determined by the actual distance between the transmit and receive antenna elements. To avoid a singularity when $d_{s,i}^{\mathrm{SI}} \to 0$, we assume that the system deployment guarantees $d_{s,i}^{\mathrm{SI}} \geq d_{\mathrm{SI}}^{\min} > 0$ for every Tx–Rx antenna pair, i.e., a minimum spacing is maintained between the transmit and receive antenna elements.

In practical full-duplex systems, SI is typically suppressed through passive isolation, analog cancellation, and digital cancellation. Instead of explicitly modeling each cancellation stage, we characterize their aggregate effect by an equivalent RSI channel. Let $0 \leq \rho_{\mathrm{SI}} \leq 1$ denote the residual SI power ratio after SI cancellation. The equivalent RSI channel is then defined as

$$\mathbf{H}_{\mathrm{RSI}} = \sqrt{\rho_{\mathrm{SI}}}\, \mathbf{H}_{\mathrm{SI}}^0. \tag{11}$$

The direct SI channel $\mathbf{H}_{\mathrm{SI}}^{0}$ depends on the transmit and receive ToMA geometries, whereas ρ_{SI} characterizes the overall SI suppression capability of the system. The matrix $\mathbf{H}_{\mathrm{RSI}}$ thus represents the equivalent residual SI channel.

C. Full-Duplex ISAC Signal Model

The main UAV simultaneously transmits multiuser DL communication signals and sensing signals through the transmit ToMA array. Define the DL communication beamforming matrix as $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_K] \in \mathbb{C}^{N_{\mathrm{T}} \times K}$, where $\mathbf{w}_k$ denotes the transmit beamformer for DL user k. Define the sensing beamforming matrix as $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_{R_{\mathrm{S}}}] \in \mathbb{C}^{N_{\mathrm{T}} \times R_{\mathrm{S}}}$, where R_{S} denotes the number of independent sensing waveforms. The combined transmit beamforming matrix is defined as

$$\mathbf{Q} = [\mathbf{W}, \mathbf{V}] \in \mathbb{C}^{N_{\mathrm{T}} \times (K + R_{\mathrm{S}})}. \tag{12}$$

Let $\mathbf{d} = [d_1^{\mathrm{D}}, \ldots, d_K^{\mathrm{D}}]^T$ denote the DL communication symbol vector, satisfying $\mathbb{E}[\mathbf{d}\mathbf{d}^H] = \mathbf{I}_K$. Let $\mathbf{z} = [z_1, \ldots, z_{R_{\mathrm{S}}}]^T$ denote the vector of independent sensing waveforms, satisfying $\mathbb{E}[\mathbf{z}\mathbf{z}^H] = \mathbf{I}_{R_{\mathrm{S}}}$. The communication symbols $\mathbf{d}$ and sensing waveforms $\mathbf{z}$ are assumed to be mutually independent. Accordingly, the full-duplex ISAC signal transmitted by the main UAV is expressed as

$$\mathbf{x} = \mathbf{W}\mathbf{d} + \mathbf{V}\mathbf{z} = \sum_{k \in \mathcal{K}} \mathbf{w}_k d_k^{\mathrm{D}} + \sum_{r=1}^{R_{\mathrm{S}}} \mathbf{v}_r z_r. \tag{13}$$

The corresponding transmit covariance matrix is

$$\mathbf{R}_x = \mathbb{E}[\mathbf{x}\mathbf{x}^H] = \mathbf{W}\mathbf{W}^H + \mathbf{V}\mathbf{V}^H = \mathbf{Q}\mathbf{Q}^H, \tag{14}$$

and hence the total transmit power at the main UAV is $\mathrm{Tr}(\mathbf{R}_x) = \|\mathbf{Q}\|_F^2$.

The signal received by DL user k consists of the desired communication signal, multiuser interference from the other DL users, the sensing signal, and receiver noise, i.e.,

$$\begin{aligned} y_k^{\mathrm{D}} = \mathbf{h}_k^H \mathbf{w}_k d_k^{\mathrm{D}} + \sum_{i \in \mathcal{K} \setminus \{k\}} \mathbf{h}_k^H \mathbf{w}_i d_i^{\mathrm{D}} \\ + \mathbf{h}_k^H \mathbf{V}\mathbf{z} + n_k, \end{aligned} \tag{15}$$

where $n_k \sim \mathcal{CN}(0, \sigma_k^2)$ denotes the additive white Gaussian noise (AWGN) at DL user k. Since the UL users have limited transmit power and the high-frequency signals considered in this paper experience severe propagation attenuation, the co-channel interference from UL users to DL users is neglected.

While transmitting the DL ISAC signal, the main UAV simultaneously receives the communication signals from the UL users and the reflected echoes from multiple sensing targets through the receive ToMA array. Let d_j^{U} denote the normalized communication symbol transmitted by UL user j, satisfying $\mathbb{E}[|d_j^{\mathrm{U}}|^2] = 1$. Furthermore, let $p_j = q_j^2$ with $q_j \geq 0$, where p_j is the transmit power of UL user j and q_j is the corresponding nonnegative transmit-amplitude variable. After aggregate SI cancellation, the signal received at the main UAV is given by

$$\mathbf{y}_{\mathrm{BS}} = \underbrace{\sum_{j \in \mathcal{J}} q_j \mathbf{f}_j d_j^{\mathrm{U}}}_{\text{UL communication signal}} + \underbrace{\sum_{l \in \mathcal{L}} \mathbf{G}_l \mathbf{x}}_{\text{target reflections}} + \underbrace{\mathbf{H}_{\mathrm{RSI}}\mathbf{x}}_{\text{residual SI}} + \mathbf{n}_{\mathrm{BS}}, \tag{16}$$

where $\mathbf{n}_{\mathrm{BS}} \sim \mathcal{CN}(\mathbf{0}, \sigma_{\mathrm{BS}}^2 \mathbf{I}_{N_{\mathrm{R}}})$ denotes the AWGN at the receive side of the main UAV.

D. Communication and Sensing Performance Metrics

For DL user k, the desired signal power is $|\mathbf{h}_k^H \mathbf{w}_k|^2$, while the remaining DL communication beams and the sensing beams are treated as interference. Hence, the SINR of DL user k is

$$\gamma_k^{\mathrm{D}} = \frac{|\mathbf{h}_k^H \mathbf{w}_k|^2}{\sum\limits_{i \in \mathcal{K} \setminus \{k\}} |\mathbf{h}_k^H \mathbf{w}_i|^2 + \|\mathbf{h}_k^H \mathbf{V}\|_2^2 + \sigma_k^2}, \qquad k \in \mathcal{K}, \tag{17}$$

where $\|\mathbf{h}_k^H \mathbf{V}\|_2^2 = \sum_{r=1}^{R_{\mathrm{S}}} |\mathbf{h}_k^H \mathbf{v}_r|^2$ represents the aggregate interference power caused by the sensing beams at DL user k. The corresponding achievable DL rate is defined as

$$R_k^{\mathrm{D}} = \log_2\left(1 + \gamma_k^{\mathrm{D}}\right). \tag{18}$$

To detect the data transmitted by UL user j, the main UAV employs a receive beamformer $\mathbf{b}_j \in \mathbb{C}^{N_{\mathrm{R}}}$ with $\|\mathbf{b}_j\|_2^2 = 1$. Since multiple target echoes and RSI are simultaneously present at the receiver of the main UAV, define the target-echo-plus-RSI interference covariance matrix for UL detection as

$$\mathbf{C}_{\mathrm{U}} = \sum_{l \in \mathcal{L}} \mathbf{G}_l \mathbf{R}_x \mathbf{G}_l^H + \mathbf{H}_{\mathrm{RSI}} \mathbf{R}_x \mathbf{H}_{\mathrm{RSI}}^H. \tag{19}$$

The first term represents the aggregate interference covariance contributed by all target echoes, while the second term represents the RSI covariance. The two types of interference are accumulated separately in the power domain, and their cross-covariance terms are neglected. Accordingly, the post-combining SINR of UL user j is

$$\gamma_j^{\mathrm{U}} = \frac{q_j^2 |\mathbf{b}_j^H \mathbf{f}_j|^2}{\sum\limits_{i \in \mathcal{J} \setminus \{j\}} q_i^2 |\mathbf{b}_j^H \mathbf{f}_i|^2 + \mathbf{b}_j^H \mathbf{C}_{\mathrm{U}} \mathbf{b}_j + \sigma_{\mathrm{BS}}^2}, \qquad j \in \mathcal{J}. \tag{20}$$

The first term in the denominator represents the multiuser interference from the other UL users, the second term represents the equivalent interference caused by target echoes and RSI, and the last term is the receiver noise. The corresponding achievable UL rate is

$$R_j^{\mathrm{U}} = \log_2\left(1 + \gamma_j^{\mathrm{U}}\right). \tag{21}$$

For sensing target l, the main UAV employs a sensing receive beamformer $\mathbf{u}_l \in \mathbb{C}^{N_{\mathrm{R}}}$ with $\|\mathbf{u}_l\|_2^2 = 1$. For point-target detection, a higher output SINR generally corresponds to better target-detection performance. Therefore, the output SINR serves as an effective and tractable metric for sensing performance. When detecting target l, the reflected echoes from the other targets and the RSI are treated as interference. Accordingly, define the corresponding interference covariance matrix as

$$\mathbf{C}_{\mathrm{S},l} = \sum_{i \in \mathcal{L} \setminus \{l\}} \mathbf{G}_i \mathbf{R}_x \mathbf{G}_i^H + \mathbf{H}_{\mathrm{RSI}} \mathbf{R}_x \mathbf{H}_{\mathrm{RSI}}^H. \tag{22}$$

Following the same treatment as in the UL communication link, the desired echo power of target l after applying $\mathbf{u}_l$ is

$\|\mathbf{u}_l^H \mathbf{G}_l \mathbf{Q}\|_2^2$. Thus, the sensing SINR corresponding to target l is

$$\gamma_l^{\mathrm{S}} = \frac{\left\|\mathbf{u}_l^H \mathbf{G}_l \mathbf{Q}\right\|_2^2}{\sum_{j\in\mathcal{J}} q_j^2 |\mathbf{u}_l^H \mathbf{f}_j|^2 + \mathbf{u}_l^H \mathbf{C}_{\mathrm{S},l} \mathbf{u}_l + \sigma_{\mathrm{BS}}^2}, \qquad l \in \mathcal{L}. \quad (23)$$

The first term in the denominator represents the interference from all UL users, the second term captures the interference caused by the echoes from the other targets and RSI, and the last term represents the receiver noise at the main UAV. To characterize the sensing performance within the same rate-based framework as the communication links, we further define the sensing-rate metric of target l as

$$R_l^{\mathrm{S}} = \log_2\left(1 + \gamma_l^{\mathrm{S}}\right). \quad (24)$$

E. Problem Formulation

Based on the above communication and sensing performance metrics, we jointly optimize the ToMA geometry, the transmit beamforming at the main UAV, the UL users' transmit powers, and the communication and sensing receive beamformers. Define the set of joint optimization variables as $\mathbf{\Xi} = \{\widetilde{\mathbf{c}}_{\mathrm{T}}, \widetilde{\mathbf{c}}_{\mathrm{R}}, \mathbf{Q}, \{\mathbf{u}_l\}_{l\in\mathcal{L}}, \{\mathbf{b}_j\}_{j\in\mathcal{J}}, \{q_j\}_{j\in\mathcal{J}}\}$. Here, $\widetilde{\mathbf{c}}_{\mathrm{T}}$ and $\widetilde{\mathbf{c}}_{\mathrm{R}}$ denote the endpoint-position vectors of the transmit and receive ToMA arrays, respectively; $\mathbf{Q}$ is the joint communication-and-sensing transmit beamforming matrix; $\mathbf{u}_l$ and $\mathbf{b}_j$ are the receive beamformers for sensing target l and UL user j, respectively; and q_j is the transmit-amplitude variable of UL user j.

The optimization objective is to maximize the weighted sum rate, given by

$$F(\mathbf{\Xi}) = \sum_{k\in\mathcal{K}} \omega_k^{\mathrm{D}} R_k^{\mathrm{D}} + \sum_{j\in\mathcal{J}} \omega_j^{\mathrm{U}} R_j^{\mathrm{U}} + \sum_{l\in\mathcal{L}} \omega_l^{\mathrm{S}} R_l^{\mathrm{S}}, \quad (25)$$

where ω_k^{D}, ω_j^{U}, and ω_l^{S} are the nonnegative performance weights assigned to DL user k, UL user j, and sensing target l, respectively, satisfying $\omega_k^{\mathrm{D}}, \omega_j^{\mathrm{U}}, \omega_l^{\mathrm{S}} \geq 0$ and $\sum_{k\in\mathcal{K}} \omega_k^{\mathrm{D}} + \sum_{j\in\mathcal{J}} \omega_j^{\mathrm{U}} + \sum_{l\in\mathcal{L}} \omega_l^{\mathrm{S}} = 1$. By adjusting these weights, the system can prioritize the relative performance of DL communication, UL communication, and target sensing. Accordingly, the joint ToMA geometry and communication–sensing resource design problem is formulated as

$$\begin{aligned}
(\mathrm{P1}) \quad & \max_{\mathbf{\Xi}} \quad F(\mathbf{\Xi}) \\
\text{s.t.} \quad & \mathrm{C1}: \|\mathbf{u}_l\|_2^2 = 1, \qquad \forall l \in \mathcal{L}, \\
& \mathrm{C2}: \|\mathbf{b}_j\|_2^2 = 1, \qquad \forall j \in \mathcal{J}, \\
& \mathrm{C3}: \|\mathbf{Q}\|_F^2 \leq P_{\mathrm{D}}^{\max}, \\
& \mathrm{C4}: 0 \leq q_j \leq \sqrt{P_{\mathrm{U},j}^{\max}}, \qquad \forall j \in \mathcal{J}, \\
& \mathrm{C5}: \|\mathbf{c}_m\|_2 = L_{\mathrm{c}}, \qquad \forall m \in \mathcal{M}, \\
& \mathrm{C6}: \|\mathbf{c}_m - \mathbf{c}_{m'}\|_2 \geq D_{\mathrm{UAV}}, \qquad \forall m \neq m'.
\end{aligned} \quad (26)$$

Constraints C1 and C2 normalize the sensing and UL receive beamformers, respectively. Constraint C3 limits the maximum total DL transmit power of the main UAV, while C4 limits the maximum transmit power of each UL user. Constraint C5 is the fixed-length cable constraint, which confines each towing-UAV endpoint to a sphere centered at the main UAV with radius L_{c}. Constraint C6 guarantees that the distance between any two towing UAVs is no smaller than the minimum safety distance D_{UAV}.

Since the communication, sensing, and SI channels are all nonlinearly coupled with the ToMA geometry, while the transmit beamformers, UL transmit powers, and receive beamformers jointly appear in multiple fractional SINR expressions, problem (P1) is a non-convex optimization problem with highly coupled variables and non-convex geometric constraints. Consequently, obtaining a globally optimal solution directly is challenging.

III. Joint ToMA Geometry and Communication–Sensing Resource Optimization

Alternating optimization (AO) is widely used to solve various optimization problems in wireless communication systems. It decomposes the original problem into several more tractable subproblems and optimizes one block of variables at a time while keeping the remaining blocks fixed. Following this principle, we decompose the original problem into two interrelated subproblems, namely, resource optimization and ToMA geometry optimization. For the resource-optimization subproblem with fixed ToMA geometry, a fractional programming (FP)-based method is employed to alternately update the communication and sensing resources. For the geometry-optimization subproblem with the current resource state fixed, a Riemannian conjugate gradient (RCG) method is used to update the endpoints of the towing UAVs. Furthermore, to alleviate the resource mismatch caused by changes in the ToMA geometry, an FP-aware candidate-evaluation mechanism is incorporated into the position-update procedure, thereby enabling joint optimization of the ToMA geometry and communication–sensing resources.

A. FP-Based Inner Resource Optimization

We adopt an FP framework to handle the optimization objective for fixed ToMA endpoint positions through the Lagrangian dual transform (LDT) and the quadratic transform (QT).

1) Receive Beamformer Update: We first fix $\mathbf{Q}$, $\{q_j\}$, and all ToMA position variables. For sensing target l, define the desired target-echo covariance and the interference-plus-noise covariance as $\mathbf{C}_l^{\mathrm{S}} = \mathbf{G}_l \mathbf{Q}\mathbf{Q}^H \mathbf{G}_l^H$ and $\mathbf{D}_l^{\mathrm{S}} = \sum_{j\in\mathcal{J}} q_j^2 \mathbf{f}_j \mathbf{f}_j^H + \mathbf{C}_{\mathrm{S},l} + \sigma_{\mathrm{BS}}^2 \mathbf{I}_{N_{\mathrm{R}}}$, respectively.

Since the sensing SINR with respect to $\mathbf{u}_l$ forms a generalized Rayleigh quotient, the maximum-SINR receive direction is given by the dominant generalized eigenvector of the matrix pair $(\mathbf{C}_l^{\mathrm{S}}, \mathbf{D}_l^{\mathrm{S}})$. Hence,

$$\mathbf{u}_l^\star = \frac{\widehat{\mathbf{u}}_l}{\|\widehat{\mathbf{u}}_l\|_2}, \qquad \widehat{\mathbf{u}}_l = \mathrm{eigvec}_{\max}\left(\mathbf{C}_l^{\mathrm{S}}, \mathbf{D}_l^{\mathrm{S}}\right). \quad (27)$$

For UL user j, define its interference-plus-noise covariance as $\mathbf{D}_j^{\mathrm{U}} = \sum_{i\in\mathcal{J}\setminus\{j\}} q_i^2 \mathbf{f}_i \mathbf{f}_i^H + \mathbf{C}_{\mathrm{U}} + \sigma_{\mathrm{BS}}^2 \mathbf{I}_{N_{\mathrm{R}}}$. The corresponding maximum-SINR/MMSE receive direction is then given by

$$\mathbf{b}_j^\star = \frac{(\mathbf{D}_j^{\mathrm{U}})^{-1} \mathbf{f}_j}{\|(\mathbf{D}_j^{\mathrm{U}})^{-1} \mathbf{f}_j\|_2}. \quad (28)$$

2) LDT and QT Auxiliary Variable Update: Define the LDT auxiliary variables as η_k^{D}, η_j^{U}, and η_l^{S}, respectively. For arbitrary $A \geq 0$ and $B > 0$, the LDT identity is

$$\log_2\left(1+\frac{A}{B}\right) = \frac{1}{\ln 2}\max_{\eta\geq 0}\Bigg[\ln(1+\eta)-\eta + (1+\eta)\frac{A}{A+B}\Bigg]. \tag{29}$$

The optimal auxiliary variable satisfies $\eta^\star = A/B = \gamma$. Therefore, in each inner iteration, the three types of LDT auxiliary variables can be directly updated as $\eta_k^{\mathrm{D}} = \gamma_k^{\mathrm{D}}$, $\eta_j^{\mathrm{U}} = \gamma_j^{\mathrm{U}}$, and $\eta_l^{\mathrm{S}} = \gamma_l^{\mathrm{S}}$. We further define the rescaled weights as $\bar{\omega}_k^{\mathrm{D}} = \omega_k^{\mathrm{D}}/\ln 2$, $\bar{\omega}_j^{\mathrm{U}} = \omega_j^{\mathrm{U}}/\ln 2$, and $\bar{\omega}_l^{\mathrm{S}} = \omega_l^{\mathrm{S}}/\ln 2$. To facilitate the subsequent QT, the total received powers of the three types of links are defined as follows. For DL user k,

$$T_k^{\mathrm{D}} = \|\mathbf{h}_k^H\mathbf{Q}\|_2^2 + \sigma_k^2. \tag{30}$$

For UL user j,

$$T_j^{\mathrm{U}} = \sum_{i\in\mathcal{J}} q_i^2|\mathbf{b}_j^H\mathbf{f}_i|^2 + \mathbf{b}_j^H\mathbf{C}_{\mathrm{U}}\mathbf{b}_j + \sigma_{\mathrm{BS}}^2. \tag{31}$$

For sensing target l,

$$T_l^{\mathrm{S}} = \|\mathbf{u}_l^H\mathbf{G}_l\mathbf{Q}\|_2^2 + \sum_{j\in\mathcal{J}} q_j^2|\mathbf{u}_l^H\mathbf{f}_j|^2 + \mathbf{u}_l^H\mathbf{C}_{\mathrm{S},l}\mathbf{u}_l + \sigma_{\mathrm{BS}}^2. \tag{32}$$

Next, QT auxiliary variables are introduced. For the DL link, let $\phi_k^{\mathrm{D}} \in \mathbb{C}$. Its optimal update is

$$\phi_k^{\mathrm{D}\star} = \frac{\mathbf{h}_k^H\mathbf{w}_k}{T_k^{\mathrm{D}}}. \tag{33}$$

For the UL link, introduce $\phi_j^{\mathrm{U}} \in \mathbb{C}$, which is updated as

$$\phi_j^{\mathrm{U}\star} = \frac{q_j\mathbf{b}_j^H\mathbf{f}_j}{T_j^{\mathrm{U}}}. \tag{34}$$

For the sensing link, since the target echo is generated by all communication and sensing streams in $\mathbf{Q}$, let $D = K + R_{\mathrm{S}}$ and introduce the vector auxiliary variable $\boldsymbol{\phi}_l^{\mathrm{S}} \in \mathbb{C}^D$. Its optimal update is

$$\boldsymbol{\phi}_l^{\mathrm{S}\star} = \frac{\mathbf{Q}^H\mathbf{G}_l^H\mathbf{u}_l}{T_l^{\mathrm{S}}}. \tag{35}$$

3) Joint Transmit Beamforming Matrix Update: With the receive beamformers, auxiliary variables, and UL transmit amplitudes fixed, define

$$\begin{aligned}
\mathbf{C}_k^{\mathrm{D}} &= \mathbf{h}_k\mathbf{h}_k^H,\\
\mathbf{C}_j^{\mathrm{U}} &= \sum_{i\in\mathcal{L}}\mathbf{G}_i^H\mathbf{b}_j\mathbf{b}_j^H\mathbf{G}_i + \mathbf{H}_{\mathrm{RSI}}^H\mathbf{b}_j\mathbf{b}_j^H\mathbf{H}_{\mathrm{RSI}},\\
\mathbf{C}_{l,\mathrm{des}}^{\mathrm{S}} &= \mathbf{G}_l^H\mathbf{u}_l\mathbf{u}_l^H\mathbf{G}_l,\\
\mathbf{C}_{l,\mathrm{int}}^{\mathrm{S}} &= \sum_{i\in\mathcal{L}\setminus\{l\}}\mathbf{G}_i^H\mathbf{u}_l\mathbf{u}_l^H\mathbf{G}_i + \mathbf{H}_{\mathrm{RSI}}^H\mathbf{u}_l\mathbf{u}_l^H\mathbf{H}_{\mathrm{RSI}}.
\end{aligned}$$

Accordingly, the negative quadratic terms with respect to $\mathbf{Q}$ can be collected as

$$\begin{aligned}
\mathbf{A}_Q = &\sum_{k\in\mathcal{K}}\bar{\omega}_k^{\mathrm{D}}(1+\eta_k^{\mathrm{D}})|\phi_k^{\mathrm{D}}|^2\mathbf{C}_k^{\mathrm{D}}\\
&+\sum_{j\in\mathcal{J}}\bar{\omega}_j^{\mathrm{U}}(1+\eta_j^{\mathrm{U}})|\phi_j^{\mathrm{U}}|^2\mathbf{C}_j^{\mathrm{U}}\\
&+\sum_{l\in\mathcal{L}}\bar{\omega}_l^{\mathrm{S}}(1+\eta_l^{\mathrm{S}})\|\boldsymbol{\phi}_l^{\mathrm{S}}\|_2^2\left(\mathbf{C}_{l,\mathrm{des}}^{\mathrm{S}}+\mathbf{C}_{l,\mathrm{int}}^{\mathrm{S}}\right),
\end{aligned} \tag{36}$$

where $\mathbf{A}_Q \succeq \mathbf{0}$.

Let $\mathbf{e}_k^{(D)}$ denote the k-th standard basis vector in $\mathbb{R}^D$, which selects the k-th DL communication beam from $\mathbf{Q}$. The corresponding linear term can be written as

$$\begin{aligned}
\mathbf{B}_Q = &\sum_{k\in\mathcal{K}}\bar{\omega}_k^{\mathrm{D}}(1+\eta_k^{\mathrm{D}})\phi_k^{\mathrm{D}}\mathbf{h}_k(\mathbf{e}_k^{(D)})^T\\
&+\sum_{l\in\mathcal{L}}\bar{\omega}_l^{\mathrm{S}}(1+\eta_l^{\mathrm{S}})\mathbf{G}_l^H\mathbf{u}_l(\boldsymbol{\phi}_l^{\mathrm{S}})^H.
\end{aligned} \tag{37}$$

After removing the terms independent of $\mathbf{Q}$, the transmit-beamforming-matrix update can be formulated as

$$\begin{aligned}
\max_{\mathbf{Q}}\quad & 2\,\mathrm{Re}\left\{\mathrm{Tr}(\mathbf{B}_Q^H\mathbf{Q})\right\} - \mathrm{Tr}(\mathbf{Q}^H\mathbf{A}_Q\mathbf{Q})\\
\text{s.t.}\quad & \|\mathbf{Q}\|_F^2 \leq P_{\mathrm{D}}^{\max}.
\end{aligned} \tag{P-Q}$$

To handle the total transmit-power constraint, introduce a Lagrange multiplier $\lambda_{\mathrm{P}} \geq 0$. From the first-order optimality condition, we obtain

$$\widehat{\mathbf{Q}} = \left(\mathbf{A}_Q + \lambda_{\mathrm{P}}\mathbf{I}_{N_{\mathrm{T}}}\right)^{-1}\mathbf{B}_Q. \tag{38}$$

If the unconstrained solution already satisfies the power constraint, we set $\lambda_{\mathrm{P}} = 0$; otherwise, the unique $\lambda_{\mathrm{P}} > 0$ is determined through a one-dimensional bisection search such that

$$\left\|(\mathbf{A}_Q + \lambda_{\mathrm{P}}\mathbf{I}_{N_{\mathrm{T}}})^{-1}\mathbf{B}_Q\right\|_F^2 = P_{\mathrm{D}}^{\max}. \tag{39}$$

The resulting $\widehat{\mathbf{Q}}$ can then be directly decomposed column-wise into the DL communication beamformers and sensing beamformers.

4) UL Transmit-Amplitude Update: With $\mathbf{Q}$ and the remaining auxiliary variables fixed, the terms related to the UL transmit amplitude q_j can be written as $2\mu_j q_j - \nu_j q_j^2 + \text{const.}$, where

$$\mu_j = \bar{\omega}_j^{\mathrm{U}}(1+\eta_j^{\mathrm{U}})\,\mathrm{Re}\{(\phi_j^{\mathrm{U}})^*\mathbf{b}_j^H\mathbf{f}_j\} \tag{40}$$

and

$$\begin{aligned}
\nu_j = &\sum_{i\in\mathcal{J}}\bar{\omega}_i^{\mathrm{U}}(1+\eta_i^{\mathrm{U}})|\phi_i^{\mathrm{U}}|^2|\mathbf{b}_i^H\mathbf{f}_j|^2\\
&+\sum_{l\in\mathcal{L}}\bar{\omega}_l^{\mathrm{S}}(1+\eta_l^{\mathrm{S}})\|\boldsymbol{\phi}_l^{\mathrm{S}}\|_2^2|\mathbf{u}_l^H\mathbf{f}_j|^2
\end{aligned} \tag{41}$$

Let $q_{j,\max} = \sqrt{P_{\mathrm{U},j}^{\max}}$. When $\nu_j > 0$, the optimal solution to this one-dimensional concave quadratic problem is

$$\widehat{q}_j = \begin{cases} 0, & \mu_j \leq 0,\\ \dfrac{\mu_j}{\nu_j}, & 0 < \mu_j < \nu_j q_{j,\max},\\ q_{j,\max}, & \mu_j \geq \nu_j q_{j,\max}. \end{cases} \tag{42}$$

Accordingly, the final transmit power of UL user j is given by $p_j = \widehat{q}_j^2$. For the degenerate case $\nu_j = 0$, the corresponding endpoint of the interval $[0, q_{j,\max}]$ is selected by maximizing the linear objective $2\mu_j q_j$.

Algorithm 1 FP-Based Inner Resource Optimization

Require: Fixed ToMA geometry C, the corresponding channels, initial feasible $\mathbf{Q}^{(0)}$ and $\{q_j^{(0)}\}$, and inner tolerance $\varepsilon_{\mathrm{in}}$, maximum inner iterations $I_{\mathrm{in}}^{\max}$.

Ensure: Optimized resource state $R^\star$.

Set $n=0$ and compute the initial true WSR $F_{\mathrm{in}}^{(0)}$.
repeat
  Update all sensing receive beamformers $\{\mathbf{u}_l^{(n+1)}\}_{l\in\mathcal{L}}$ according to (27).
  Update all UL receive beamformers $\{\mathbf{b}_j^{(n+1)}\}_{j\in\mathcal{J}}$ according to (28).
  Compute the current true SINRs and update the LDT variables $\{\eta_k^{\mathrm{D}}\}$, $\{\eta_j^{\mathrm{U}}\}$, and $\{\eta_l^{\mathrm{S}}\}$.
  Compute $\{T_k^{\mathrm{D}}\}$, $\{T_j^{\mathrm{U}}\}$, and $\{T_l^{\mathrm{S}}\}$, and update the QT variables $\{\phi_k^{\mathrm{D}}\}$, $\{\phi_j^{\mathrm{U}}\}$, and $\{\boldsymbol{\phi}_l^{\mathrm{S}}\}$.
  Construct $\mathbf{A}_Q$ and $\mathbf{B}_Q$ according to (36) and (37).
  Determine λ_{P} by one-dimensional bisection when necessary, and update $\mathbf{Q}^{(n+1)}$ according to (38).
  Compute $\{\mu_j,\nu_j\}_{j\in\mathcal{J}}$ and update all UL transmit amplitudes $\{q_j^{(n+1)}\}_{j\in\mathcal{J}}$ according to (42).
  Recompute the true SINRs and the true WSR $F_{\mathrm{in}}^{(n+1)}$.
  Set $n\leftarrow n+1$.
until $\dfrac{|F_{\mathrm{in}}^{(n)}-F_{\mathrm{in}}^{(n-1)}|}{\max\{1,|F_{\mathrm{in}}^{(n-1)}|\}}\le\varepsilon_{\mathrm{in}}$ or $n\ge I_{\mathrm{in}}^{\max}$
Set $R^\star=\{\mathbf{Q}^{(n)},\{q_j^{(n)}\},\{\mathbf{u}_l^{(n)}\},\{\mathbf{b}_j^{(n)}\}\}$.
return $R^\star$.

5) Inner Resource Alternating Update: For fixed ToMA geometry, the inner resource optimization sequentially updates the sensing receive beamformers, UL receive beamformers, LDT auxiliary variables, QT auxiliary variables, joint transmit matrix, and UL transmit amplitudes. After each resource update, the true SINRs and the true WSR are recomputed, and the above procedure is repeated until the prescribed inner stopping criterion is satisfied.

B. FP-Aware RCG-Based ToMA Position Optimization

We employ an RCG method to sequentially update the ToMA endpoints over the spherical manifolds while enforcing the collision-avoidance condition to preserve geometric feasibility. All transmit and receive endpoints are updated in a prescribed order. At the beginning of each endpoint update, let the current complete ToMA geometry and current resource state be denoted by C_t and R_t, respectively, where R_t contains $\mathbf{Q}$, $\{q_j\}$, $\{\mathbf{u}_l\}$, and $\{\mathbf{b}_j\}$. During the position-gradient computation and RCG search-direction construction for the current endpoint, the resource state R_t is kept fixed. Therefore, the position gradient used in this work is $g_t=\mathrm{grad}_C F(C_t,R_t)$, i.e., the ToMA geometry gradient under the fixed current resource state, rather than the gradient of the exact reduced objective $\Phi(C)=F(C,R^\star(C))$. The limited FP updates at a candidate position are used only for the subsequent candidate evaluation.

1) Riemannian Gradient Computation: Since constraint C5 requires each endpoint to satisfy $\|\mathbf{c}_m\|_2=L_{\mathrm{c}}$, the feasible position of the m-th towing UAV lies on the spherical manifold $\mathcal{S}_m=\{\mathbf{c}\in\mathbb{R}^3:\mathbf{c}^T\mathbf{c}=L_{\mathrm{c}}^2\}$. At the current endpoint $\mathbf{c}_m$, the tangent space is $T_{\mathbf{c}_m}\mathcal{S}_m=\{\mathbf{v}\in\mathbb{R}^3:\mathbf{c}_m^T\mathbf{v}=0\}$. Hence, the Euclidean gradient of the WSR with respect to $\mathbf{c}_m$ is first computed and then orthogonally projected onto the tangent space, yielding the Riemannian gradient $\mathbf{g}_m=(\mathbf{I}_3-\mathbf{c}_m\mathbf{c}_m^T/L_{\mathrm{c}}^2)\nabla_{\mathbf{c}_m}F$.

Since the ToMA endpoint positions affect the communication, sensing, and RSI channels through the antenna-element coordinates, deriving an analytical expression for the gradient of F with respect to $\mathbf{c}_m$ is cumbersome. Therefore, we approximate the Euclidean gradient using a central finite difference. For the a-th coordinate, $a\in\{1,2,3\}$,

$$[\nabla_{\mathbf{c}_m}F]_a\approx\frac{F(\mathbf{c}_m+\delta\mathbf{e}_a,R_t)-F(\mathbf{c}_m-\delta\mathbf{e}_a,R_t)}{2\delta},\quad(43)$$

where $\delta>0$ is the finite-difference step size and $\mathbf{e}_a$ is the a-th standard basis vector in $\mathbb{R}^3$. For each function evaluation, the antenna-element coordinates and the associated channels are regenerated according to the perturbed endpoint, while the resource variables remain fixed at the current R_t. These perturbed points are used only for gradient estimation and are not treated as admissible ToMA positions.

2) RCG Search Direction and Spherical Retraction: To exploit the directional information from the previous position update, we adopt a Polak–Ribiere+ (PR+) RCG search direction, given by

$$\boldsymbol{\psi}_m^{(t)}=\mathbf{g}_m^{(t)}+\beta_m^{(t)}\mathcal{T}\left(\boldsymbol{\psi}_m^{(t-1)}\right),\quad(44)$$

where $\mathcal{T}(\cdot)$ denotes the vector transport that maps the tangent vector from the previous tangent space to the current one. The PR+ coefficient is $\beta_m^{(t)}=\max\{0,\langle\mathbf{g}_m^{(t)},\mathbf{g}_m^{(t)}-\mathcal{T}(\mathbf{g}_m^{(t-1)})\rangle/\|\mathbf{g}_m^{(t-1)}\|_2^2\}$. If the resulting direction does not guarantee ascent, i.e., $\langle\mathbf{g}_m^{(t)},\boldsymbol{\psi}_m^{(t)}\rangle\le0$, the search direction is restarted as $\boldsymbol{\psi}_m^{(t)}=\mathbf{g}_m^{(t)}$, thereby ensuring an ascent direction for the subsequent search.

Since directly moving along a tangent direction generally leaves the spherical manifold, a retraction is used to map the trial point back onto the sphere. Given a trial step size $\tau_m>0$, the candidate endpoint is constructed as

$$\mathbf{c}_m^{\mathrm{cand}}(\tau_m)=L_{\mathrm{c}}\frac{\mathbf{c}_m+\tau_m\boldsymbol{\psi}_m}{\|\mathbf{c}_m+\tau_m\boldsymbol{\psi}_m\|_2}.\quad(45)$$

The normalization directly gives $\|\mathbf{c}_m^{\mathrm{cand}}\|_2=L_{\mathrm{c}}$, so C5 is automatically satisfied at each position update. In addition, the candidate position must satisfy the collision-avoidance constraint $\|\mathbf{c}_m^{\mathrm{cand}}-\mathbf{c}_{m'}\|_2\ge D_{\mathrm{UAV}}$ for all $m'\neq m$. If this condition is violated, the current candidate is rejected and the trial step size is reduced to generate a new candidate.

3) FP-Aware Candidate Evaluation and Armijo Backtracking: Conventional fixed-resource Armijo backtracking directly evaluates whether a candidate geometry improves the objective using $F(C_{\mathrm{cand}},R_t)$. In the considered system, however, changing the ToMA position simultaneously alters the DL, UL, sensing, and RSI channels. Consequently, the resources obtained from the previous geometry may be poorly matched to the new position, causing a potentially useful candidate to be rejected or leading to an excessively small step size.

To mitigate this geometry–resource mismatch, we adopt an FP-aware candidate-evaluation strategy. For a geometrically feasible candidate C_{cand}, the corresponding ToMA geometry and all associated channels are reconstructed. Using the current resource state R_t as a warm start, at most two inner FP iterations are then performed under the candidate channels. Denote the resulting candidate resource state by $\widetilde{R}_{\mathrm{cand}} = \mathcal{T}_{\mathrm{FP}}^{(2)}(C_{\mathrm{cand}}; R_t)$, where the superscript (2) indicates that the maximum number of FP iterations is two; the update can terminate earlier if the inner stopping criterion is already satisfied.

The FP-aware candidate objective is then defined as

$$\widehat{\Phi}_2(C_{\mathrm{cand}}; R_t) = F(C_{\mathrm{cand}}, \widetilde{R}_{\mathrm{cand}}). \tag{46}$$

It should be emphasized that $\widehat{\Phi}_2$ is only a candidate-geometry performance measure obtained from a finite number of warm-start FP updates and is not identical to the exact reduced objective obtained by fully optimizing the resource variables.

The candidate position is accepted according to the following FP-aware Armijo ascent condition:

$$\widehat{\Phi}_2(C_{\mathrm{cand}}; R_t) \geq F(C_t, R_t) + \mu_{\mathrm{A}} \tau_m \langle \mathbf{g}_m, \boldsymbol{\psi}_m \rangle, \quad 0 < \mu_{\mathrm{A}} < 1. \tag{47}$$

If the above condition is satisfied, the candidate position and its FP-readapted resource state are accepted simultaneously, i.e., $C_{t+1} = C_{\mathrm{cand}}$ and $R_{t+1} = \widetilde{R}_{\mathrm{cand}}$. Thus, subsequent endpoint updates do not combine the newly accepted position with the old resource state R_t; instead, the new pair (C_{t+1}, R_{t+1}) is directly used as the current state. If either the collision-avoidance condition or the FP-aware Armijo condition is violated, the step size is reduced according to $\tau_m \leftarrow \zeta_{\mathrm{out}} \tau_m$, where $0 < \zeta_{\mathrm{out}} < 1$, and the spherical retraction and candidate evaluation are repeated. If no acceptable candidate is found after the prescribed maximum number of backtracking steps, the current geometry C_t and resource state R_t are retained. During candidate-position evaluation, the communication, sensing, and RSI channels are all recomputed according to the new geometry. In particular, $\mathbf{H}_{\mathrm{RSI}}$ varies with the distances between the transmit and receive ToMA elements and must therefore be updated for each candidate position; however, it affects candidate performance only through the UL and sensing WSR terms and is not used as an additional feasibility test.

C. Overall Joint Optimization Procedure

Combining the above inner resource optimization and outer ToMA geometry optimization, we construct a joint iterative algorithm to solve problem (P1). First, multiple feasible initial ToMA endpoint configurations satisfying the fixed cable-length and minimum inter-UAV safety-distance constraints are generated, and the subsequent joint optimization is performed independently for each initialization. Based on each initial ToMA endpoint configuration, the joint transmit beamforming matrix $\mathbf{Q}$ and the UL transmit amplitudes $\{q_j\}$ are initialized while satisfying the total DL transmit-power constraint and the corresponding UL power constraints.

Algorithm 2 FP-Aware RCG-Based ToMA Position Optimization

Require: Current ToMA geometry C, resource state R, finite-difference step δ, initial step size τ_0, Armijo parameter μ_{A}, backtracking factor ζ_{out}, and maximum backtracking iterations I_{A}.
Ensure: Updated ToMA geometry C^+ and resource state R^+.

Set $(C_t, R_t) \leftarrow (C, R)$.
for each endpoint m in the prescribed Tx–Rx update order **do**
  Compute $\nabla_{\mathbf{c}_m} F(C_t, R_t)$ using the central finite difference in (43), with R_t fixed.
  Project the Euclidean gradient onto $T_{\mathbf{c}_m}\mathcal{S}_m$ to obtain the Riemannian gradient $\mathbf{g}_m$.
  If no previous RCG state is available for endpoint m, set $\boldsymbol{\psi}_m = \mathbf{g}_m$; otherwise, construct the PR+ RCG direction according to (44). Restart with $\boldsymbol{\psi}_m = \mathbf{g}_m$ if $\langle \mathbf{g}_m, \boldsymbol{\psi}_m \rangle \leq 0$.
  Set $\tau_m \leftarrow \tau_0$ and accepted $\leftarrow$ false.
  for $r = 1, \ldots, I_{\mathrm{A}}$ **do**
    Generate $\mathbf{c}_m^{\mathrm{cand}}(\tau_m)$ by the spherical retraction in (45) and form the complete candidate geometry C_{cand}.
    if C_{cand} violates the collision-avoidance constraint C6 **then**
      $\tau_m \leftarrow \zeta_{\mathrm{out}} \tau_m$.
    else
      Reconstruct all candidate channels and, using R_t as a warm start, perform at most two FP update iterations to obtain $\widetilde{R}_{\mathrm{cand}} = \mathcal{T}_{\mathrm{FP}}^{(2)}(C_{\mathrm{cand}}; R_t)$.
      Evaluate $\widehat{\Phi}_2(C_{\mathrm{cand}}; R_t)$ according to (46).
      if the FP-aware Armijo condition (47) is satisfied **then**
        $(C_t, R_t) \leftarrow (C_{\mathrm{cand}}, \widetilde{R}_{\mathrm{cand}})$.
        accepted $\leftarrow$ true;
        break.
      else
        $\tau_m \leftarrow \zeta_{\mathrm{out}} \tau_m$.
      end if
    end if
  end for
  if accepted $=$ false **then**
    Keep (C_t, R_t) unchanged.
  end if
end for
Set $(C^+, R^+) \leftarrow (C_t, R_t)$.
return (C^+, R^+).

In each joint iteration, the current ToMA geometry is first fixed and the inner FP-based resource optimization is performed. After the inner resource update is completed, the current resource state together with the current ToMA geometry serves as the starting point for the outer position optimization.

The FP-aware RCG method is then used to sequentially update the ToMA endpoints. Riemannian optimization, the RCG search direction, and spherical retraction are first used to generate candidate positions. For each geometrically feasible

Algorithm 3 Overall Joint Alternating Optimization

Require: System parameters, number of initializations N_{start}, outer tolerance $\varepsilon_{\mathrm{out}}$, and maximum outer iterations $I_{\mathrm{out}}^{\max}$.
Ensure: Optimized ToMA geometry $C^{\star}$ and resource state $R^{\star}$.

```
Generate N_start feasible initial geometries {C_s^(0)}_{s=1}^{N_start} satisfying C5 and C6.
for s = 1, ..., N_start do
    Initialize a feasible resource state R_s^(0) for C_s^(0), and compute the initial true WSR F_s^(0).
    Set t = 0 and c_s = 0.
    repeat
        Run Algorithm 1 with fixed C_s^(t), initialized by Q_s^(t) and {q_{j,s}^(t)}, to obtain R̂_s^(t).
        Apply Algorithm 2 to (C_s^(t), R̂_s^(t)) and obtain (C_s^(t+1), R_s^(t+1)).
        Reconstruct all channels and compute the true WSR F_s^(t+1).
        Compute Δ_s^(t) = |F_s^(t+1) − F_s^(t)| / max{1, |F_s^(t)|}.
        if Δ_s^(t) ≤ ε_out then
            c_s ← c_s + 1.
        else
            c_s ← 0.
        end if
        t ← t + 1.
    until c_s = 2 or t ≥ I_out^max
    Store the final solution (C_s^⋆, R_s^⋆) and its true WSR F_s^⋆.
end for
Select s^⋆ = arg max_s F_s^⋆.
Set (C^⋆, R^⋆) ← (C_{s^⋆}^⋆, R_{s^⋆}^⋆).
return (C^⋆, R^⋆).
```

candidate, the corresponding communication, sensing, and RSI channels are reconstructed, and at most two inner FP iterations are performed using the current resource state as a warm start. The readapted true joint WSR is then used for FP-aware Armijo acceptance. If the acceptance condition is satisfied, the candidate position and the corresponding candidate resource state are accepted simultaneously and are used as the current state for updating the next endpoint.

After all transmit and receive ToMA endpoints have been updated, all channels, communication and sensing SINRs, and the true WSR corresponding to the current geometry are recomputed before proceeding to the next joint iteration. To avoid premature termination caused by occasional small fluctuations of the objective value near convergence, the joint optimization for the current initialization is terminated when the relative WSR change satisfies

$$\frac{|F^{(t+1)} - F^{(t)}|}{\max\{1, |F^{(t)}|\}} \leq \varepsilon_{\mathrm{out}} \tag{48}$$

for two consecutive outer iterations.

D. Computational Complexity Analysis

The main computational cost of the overall algorithm arises from the matrix decompositions involved in the receive-beamformer updates and the joint transmit-beamforming matrix update in Algorithm 1, as well as the repeated objective-function evaluations based on the central finite difference and the warm-start FP updates for candidate positions during the FP-aware Armijo backtracking procedure in Algorithm 2.

Algorithm 1 performs the inner resource optimization for a fixed ToMA geometry. Its main computational cost arises from the sensing and UL receive-beamformer updates and the joint transmit-beamforming matrix update. Let I_{in} denote the number of inner FP iterations, I_{λ} denote the total number of trial values of λ_{P} evaluated during the bracketing and bisection procedures, and define $D = K + R_{\mathrm{S}}$. Let C_{FP} denote the dominant computational cost of one FP update. It is given by $C_{\mathrm{FP}} = \mathcal{O}\big((L+J)N_{\mathrm{R}}^3 + N_{\mathrm{T}}^3 + I_{\lambda}(N_{\mathrm{T}}^3 + N_{\mathrm{T}}^2 D) + J(J+L)N_{\mathrm{R}}\big)$. Accordingly, the computational complexity of Algorithm 1 is $\mathcal{O}_1 = \mathcal{O}(I_{\mathrm{in}} C_{\mathrm{FP}})$.

Algorithm 2 performs the FP-aware RCG-based optimization of the ToMA positions. Since the gradient of each three-dimensional ToMA endpoint is evaluated using the central finite difference, approximately six fixed-resource objective-function evaluations are required for each endpoint. In addition, at most two warm-start FP updates are performed for each Armijo candidate. Let C_F denote the computational cost of one fixed-resource objective-function evaluation and I_{A} denote the maximum number of Armijo backtracking steps for each endpoint. The computational complexity of Algorithm 2 is therefore $\mathcal{O}_2 = \mathcal{O}(6MC_F + MI_{\mathrm{A}}(C_F + 2C_{\mathrm{FP}}))$.

Algorithm 3 alternately executes Algorithms 1 and 2 for multiple feasible initial ToMA geometries and selects the feasible solution with the largest final WSR. Let N_{start} denote the number of initializations and I_{out} denote the number of joint alternating-optimization iterations for each initialization. Its overall computational complexity is

$$\begin{aligned}\mathcal{O}_3 = \mathcal{O}\Big(N_{\mathrm{start}} I_{\mathrm{out}} \big[& I_{\mathrm{in}} C_{\mathrm{FP}} + 6MC_F \\ & + MI_{\mathrm{A}}(C_F + 2C_{\mathrm{FP}})\big]\Big).\end{aligned} \tag{49}$$